LP RISK ENGINE

Single-Pool Overview

A risk-first framework for concentrated liquidity deployment

ETH/USDC 0.05% | Ethereum

Uniswap v3 | Weekly decision

THE CORE QUESTION

Have similar historical market and pool environments tended to compensate LPs adequately for the price-path risk they took in the following week?

1. Decision lens

Fee generation is not the same as fee adequacy.

A concentrated-liquidity LP can earn attractive fees while still suffering an unfavourable outcome when inventory drift and rebalancing-related costs overwhelm those fees.

FEE COVERAGE RATIO (FCR)

FCR = fees earned / modelled rebalancing costs

Above 1: fees covered modelled costs

2. How the engine works

It turns a complex on-chain LP problem into an interpretable weekly decision process.

1 Observe

Recent price path and pool state

Price signals: net return, displacement, path length, balance and shock behaviour.

On-chain state: TVL, ticks, trading activity and liquidity distribution.

→

2 Reconstruct

LP economics across range choices

Models a standardised $10,000 position at 5%, 10% and 20% ranges.

Estimates trading-fee revenue and price-path-related rebalancing costs.

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3 Compare

Similar historical environments

Classifies the current week into an interpretable regime, then examines next-week outcomes from comparable historical weeks.

3. Weekly output

A deployment judgement, not a prediction of ETH price.

DEPLOY OR WAIT

Whether historical evidence supports deployment under current conditions.

RANGE CHOICE

The most suitable candidate range if deployment is justified.

FRAGILITY WARNINGS

Weak FCR, downside risk, or transition-risk signals.

4. Reliability and guardrails

No look-ahead

Only information available at the end of the completed week is used.

Interpretable by design

Regime labels link recommendations to understandable market behaviour.

Context over ratio alone

FCR is reviewed with fee value, costs, net surplus and downside outcomes.

Not a price forecast

The framework assesses historical conditional risk, not exact future price.

Purpose: Assess whether LP fee revenue has historically been sufficient to compensate for the risk path.

Not: A profit guarantee, a generic alpha model, or a precise ETH price forecast.